# Tutorial-3

A retail company wants to predict monthly sales revenue based on marketing and operational factors.

Input Features:

-> Advertising spend ₹ in thousands

-> Number of salespersons.

-> Store footfall (Per month)

-> Discount Rate (%)

-> Target Variable:

-> Monthly Sales Revenue (₹ in Lakhs)

| Ad spend | Salesperson | Football | Discount (%) | Sales Revenue |
|---|---|---|---|---|
| 50 | 5 | 1,200 | 10 | 8.5 |
| 70 | 6 | 1,500 | 12 | 10.2 |
| 40 | 4 | 1,000 | 8 | 6.8 |
| 90 | 8 | 1,900 | 15 | 14.0 |
| 60 | 6 | 1,400 | 10 | 9.5 |
| 80 | 7 | 1,700 | 14 | 12.5 |
| 55 | 5 | 1,300 | 9 | 8.9 |
| 100 | 9 | 2,100 | 18 | 16.0 |
| 65 | 6 | 1,450 | 11 | 9.8 |
| 45 | 4 | 1,100 | 8 | 7.2 |

1) Define the dependent variables.

Dependent variables is monthly sales Revenue bcoz MSR depends on other factors like advertising spend, Number of salespersons, store football and discount rate.

2. Write the Multiple Linear Regression Model.

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Where $Y$ = Sales Revenue

$X_1$ = Advertising Spend.

$X_2$ = Number of Sales Person

$X_3$ = Store football

$X_4$ = Discount rate

$\beta_0$ = Intercept

$\beta_1, \beta_2, \beta_3, \beta_4$ = Coefficients

3. Final Model:

Sales = $\beta_0 + \beta_1$ (Ad Spend) + $\beta_2$ (Sales persons) + $\beta_3$ (football) + $\beta_4$ (Discount)

Substitute each row

Row 1 - $8.5 = \beta_0 + 50\beta_1 + 5\beta_2 + 1200\beta_3 + 10\beta_4$

Row 2 - $10.2 = \beta_0 + 70\beta_1 + 4\beta_2 + 1500\beta_3 + 12\beta_4$

Row 3 - $6.8 = \beta_0 + 40\beta_1 + 4\beta_2 + 1000\beta_3 + 8\beta_4$

Row 4 - $14 = \beta_0 + 90\beta_1 + 8\beta_2 + 1900\beta_3 + 15\beta_4$

Row 5 - $9.5 = \beta_0 + 60\beta_1 + 6\beta_2 + 1400\beta_3 + 10\beta_4$

Row 6 - $12.5 = \beta_0 + 80\beta_1 + 7\beta_2 + 1700\beta_3 + 14\beta_4$

Row 7 - $8.9 = \beta_0 + 55\beta_1 + 5\beta_2 + 1300\beta_3 + 9\beta_4$

Row 8 - $16.0 = \beta_0 + 100\beta_1 + 9\beta_2 + 2100\beta_3 + 18\beta_4$

Row 9 - $9.8 = \beta_0 + 65\beta_1 + 6\beta_2 + 1400\beta_3 + 11\beta_4$

Row 10 - $7.2 = \beta_0 + 45\beta_1 + 4\beta_2 + 1100\beta_3 + 8\beta_4$

Discuss the effect of discount rate on sales growth revenue

Discount effort:

Discount = 8, Sales = 6.8, discount = 18, Sales = 16

Since $18 > 8 \Rightarrow 16 > 6.8$

Sales increase with discount (Positive effect)

4. Find MSE for 5 iterations

$$MSE = \frac{1}{N}\sum_{i=1}^{N}(y - \hat{y}_i)^2 \quad n = 10$$

$\hat{y}_i = [8, 9, 7, 13, 10, 11, 9, 15, 10, 8]$

Square errors:

$(0.15)^2 + (1.2)^2 + (-0.2)^2 + 1^2 + (-0.5)^2 + (-0.1)^2 + 1^2 + (-0.2)^2 + (0.8)^2$

Sum = 6.92

$$MSE_1 = \frac{6.92}{10} = 0.692$$

Iteration 2:

$\hat{y}_2 = [8.3, 9.8, 6.9, 13.6, 9.7, 12, 8.8, 15.5, 9.9, 7.5]$

Sum = 2.01

$$MSE_2 = \frac{2.01}{10} = 0.201$$

Iteration 3:

$\hat{y}_3 = [8.4, 10, 6.85, 13.8, 9.6, 12.3, 8.85, 15.8, 9.85, 7.35]$

Sum of squared errors = 0.062

$$MSE_3 = \frac{0.62}{10} = 0.062$$

Iteration 4:

$\hat{y}_4 = [8.45, 10.1, 6.82, 13.9, 9.55, 12.4, 8.88, 15.9, 9.82, 7.28]$

Sum of squared errors = 0.18

$$MSE_4 = \frac{0.18}{10} = 0.018$$

Iteration 5:

$\hat{y}_5 = [8.5, 10.21, 6.8, 14, 9.5, 12.5, 8.9, 16, 9.8, 7.2]$

All errors = 0

$MSE_5 = 0$